

Karim Ben Boubaker

AI Engineer | AI Agents | Information Retrieval | Knowledge Graphs

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Open to relocation

Summary

AI Engineer experienced in developing and evaluating large language model applications, including retrieval-augmented generation (RAG), tool-using agents, and knowledge-graph-based retrieval. Background spans healthcare, legal technology, and regulatory compliance, with hands-on work in Python backend development and cloud deployment.

Experience

AI Research Engineer Intern | Hochschule Offenburg & Hahn-Schickard | Offenburg, Germany March 2026 – September 2026

- Built **PydanticAI** agents using **agentic RAG** to retrieve clinical evidence and estimate conditional probabilities for **Bayesian network parameterization**.
- Designed **tree-structured CPTs** (conditional probability tables) based on **context-specific independence**, reducing **LLM calls by 79% on average**.
- Developed an **LLM evaluation framework using Langfuse** to compare **four large language models (LLMs) across 216 runs**, measuring expert agreement, inter-model agreement, and run-to-run variability.
- Built a **React interface** for **human-in-the-loop clinical review** of probability estimates, tree structures, and supporting medical evidence.

AI Engineer (Part-time) | GreenEarthX | Palo Alto, CA (Remote)

November 2025 – March 2026

- Built a **RAG workflow using LangGraph** for regulatory compliance assessment, achieving **88% accuracy** on **100 expert-validated scenarios**.
- Designed a **Neo4j knowledge graph** of EU regulations, modeling amendments, citations, and cross-document dependencies to support **multi-hop retrieval** of regulatory evidence.
- Deployed containerized **FastAPI services** on **AWS ECS** and integrated **Langfuse** to trace LLM execution, monitor token usage, and troubleshoot production issues.

AI Engineer (Part-time) | Oratio Technologies | Tunis, Tunisia

June 2025 – November 2025

- Reduced average **time to first token (TTFT) from 15 to 5 seconds** for a legal RAG system by executing asynchronous tasks concurrently and reusing database clients and connection pools.
- Designed and implemented an **event-driven ETL pipeline** on **Azure** using **Terraform**, processing **30,000+ crawled documents** asynchronously into **MongoDB** and **Azure Cosmos DB**.
- Built synthetic evaluation datasets and an **LLM-as-a-judge evaluation framework** integrated into **CI/CD** to detect RAG quality regressions before deployment.

ML Engineering Intern | ACTIA Engineering Services | Ariana, Tunisia

June 2024 – September 2024

- Designed, trained, and optimized a **convolutional neural network (CNN)** for speech command recognition using **TensorFlow**, targeting low-latency inference on automotive microcontrollers.

Projects

Ckeeper: AI Agent for Cloud Infrastructure | [GitHub Repository](#)

May 2025 – November 2025

- Built a **plan-and-execute agent** that translated natural-language infrastructure requests into executable workflows using GCP, Terraform, and GitHub.
- Contributed to product definition, MVP development, and pitch presentations; the team placed in the **top 5 of 30 teams** in the OSTX Bootcamp Ideation Program (*Open Startup Tunisia*).

GPT-2-Style Language Model from Scratch | [Technical Blog Post](#)

September 2025 – December 2025

- Implemented and trained a **35M-parameter GPT-2-style model** in **PyTorch**, including a custom tokenizer, pretraining, and instruction fine-tuning; published a technical walkthrough.

Education

Engineering Diploma (Master's Equivalent) – Industrial Computing and Automation

National Institute of Applied Sciences and Technology (INSAT) | Tunis, Tunisia

September 2021 – September 2026

Skills & Technologies

Programming Languages: Python, TypeScript, SQL, Bash, C, C++

Frontend: React

AI / LLM Engineering: RAG, AI Agents, LLM Evaluation, LangChain, LangGraph, Langfuse, PydanticAI, LlamaIndex, OpenRouter

ML Engineering: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers

Backend / Data Engineering: FastAPI, Django, SQLAlchemy, MongoDB, PostgreSQL, Neo4j, Qdrant

Cloud / DevOps: Azure, AWS, Docker, Terraform, CI/CD, GitHub, GitLab, GitHub Actions

Certifications: [AWS Certified Cloud Practitioner](#), [NVIDIA: Building RAG Agents with LLMs \(Jan 2025\)](#)

Languages: English (C1, [TOEFL 110](#)), French (B2), German (A2), Arabic (Native)